

PRATHAM VIDHANI

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PROFESSIONAL SUMMARY

Software Engineer specializing in Java backend systems and Python AI/ML pipelines, with production experience building reactive API gateways (Spring Boot, Redis, Resilience4J), real-time distributed systems (WebSocket, Redis Pub/Sub), and RAG-based document intelligence (LangChain, pgvector). Co-authored an IEEE publication on deep learning applications and solved 300+ Data Structures & Algorithms (DSA) problems in C++.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS, Javascript

Frameworks & Libraries: Spring Boot, FastAPI, React

Backend Concepts: REST APIs, JWT, Microservices

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI / ML: TensorFlow, PyTorch, LLMs, LangChain, RAG Pipelines

Cloud & DevOps: Docker, CI/CD, AWS

Tools & Practices: Git, GitHub, Agile/Scrum

WORK EXPERIENCE

Software Engineer Intern | Tektutor

Feb 2026 – Jul 2026

- Built a cross-platform desktop invoicing application in C++ and Qt6, replacing a manual spreadsheet workflow and cutting invoice processing time by an estimated 40%.
- Engineered an event-driven tax and computation system across 6 normalized SQLite tables, enabling instant invoice recalculation and cutting invoice generation time by an estimated 60%.
- Built a PDF generation pipeline supporting 1–50+ line items per invoice, delivering a fully offline-capable billing solution for the business.

AI Operations Engineer | Amari AI

Sep 2025 – Jan 2026

- Debugged and resolved 80–100 daily pipeline errors in a production AI document-extraction system, sustaining 99%+ data accuracy across 1,000+ documents processed weekly.
- Cut recurring pipeline failures by 90%+ through root-cause analysis, directly reducing manual rework and downstream data-quality escalations for the AI engineering team.
- Standardized error-resolution workflows across the SDLC by authoring 15+ SOPs and leading GitHub PR reviews, cutting onboarding time for new team members handling pipeline incidents.

PROJECTS

Reactive API Gateway · Java 17 · Spring Boot 3 · Spring Cloud Gateway · Project Reactor · Redis · Resilience4j · Netty · Docker

- Architected a reactive API Gateway using Spring Cloud Gateway and Netty, enforcing per-route rate limits up to 20 req/sec via Redis-backed policies.
- Built a non-blocking observability pipeline capturing 7 metadata fields per request, using Redis sliding windows for real-time traffic analysis.
- Implemented fault-tolerant request handling with Resilience4j circuit breakers, exporting metrics across 5 endpoints with sub-15-second failure detection.

Real-Time Collaborative Task Board · Java 17 · Spring Boot · WebSocket · Redis · PostgreSQL · React · Docker · Microservices

- Architected a real-time Kanban board using Spring Boot and WebSocket, syncing card and column updates live across 10 concurrent users with sub-100ms latency.
- Built a Redis pub/sub layer to broadcast state across 3 containerized backend instances, resolving race conditions from simultaneous card drags.
- Built a React drag-and-drop frontend backed by JWT-secured REST APIs and PostgreSQL, persisting board and card state across 15+ boards.

Document Intelligence API · FastAPI · LangChain · pgvector · PostgreSQL · Redis · Docker · OpenAI API · AWS Cloud

- Built a Retrieval-Augmented Generation (RAG) pipeline with LangChain and pgvector, enabling semantic search with source citations across 500+ documents.
- Cut LLM inference costs by 60%+ by building a Redis-based semantic cache that serves repeat queries without re-calling the model.
- Deployed a 3-service FastAPI backend via Docker to AWS EC2, exposing 3 JWT-secured REST APIs with rate limiting and monitoring.

EDUCATION

B. Tech – Computer Science and Engineering

Sep 2021 – Aug 2025

Vellore Institute of Technology, Vellore. CGPA: 7.8 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming (OOP), Computer Networks

ACHIEVEMENTS & PUBLICATIONS

- Co-authored IEEE publication applying transfer learning and CNN architectures (VGG, ResNet, InceptionV3) to crop water estimation, achieving 90% accuracy improvement over baseline models. [\[IEEE Publication\]](#)
- Solved 300+ Data Structures & Algorithms (DSA) problems on LeetCode and Codeforces in C++, covering dynamic programming and object-oriented programming (OOP) principles. [\[LeetCode\]](#)
- Led PR & outreach for a 10-member team at Becoming I Foundation (BIF), increasing audience reach by 40%+ across 5+ college events.